FERNANDES K J

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SUMMARY

Software Development Engineer with 2+ years of experience building secure, high-throughput **Android SDKs** and **backend systems** for fintech and payments. Built BillDesk's **Android Payment Gateway SDK** with **Kotlin, Jetpack Compose** and **MVVM/MVI**, and backend services with **Java, Spring Boot, Kafka** and **REST APIs**. Strong in payment flows (UPI, NetBanking,Cards) and understanding of **payment flows, application security, event-driven architecture, and performance optimization**, with a focus on building reliable, production-ready solutions.

EXPERIENCE

Software Development Engineer — Hatio Innovations (BillDesk Subsidiary) *Aug 2024 – Present*

Tech Stack: Android (Jetpack Compose, MVVM, MVI), Vue.js, React, Spring Boot, Kafka, REST APIs, Kotlin, Java

BillDesk Payment Gateway SDK :

- Built BillDesk's **Android Payment Gateway SDK** from the ground up, a core payment integration platform enabling merchants to process transactions within an ecosystem handling **3M+ transactions daily** across **200+ billers**.
- Designed and developed the SDK using **MVVM + MVI architecture**, delivering a more performant and maintainable codebase.
- Own the **payment methods module**, having implemented **Cards, NetBanking 2.0, UPI Intent, and BHIM UPI** integrations, plus the **One-Time Mandate flow built from scratch**, expanding the SDK's transaction capabilities.
- Improved SDK rendering performance by **21.38% (**derivedState,GraphicLayer, class stability etc **)**while concurrently shipping new payment method integrations, balancing speed and feature velocity in a production system.
- Engineered critical security infrastructure including **Root detection, SSL pinning, anti-tampering (Frida) detection, and a base-URL enforcement interceptor** hardening the SDK against common mobile payment vulnerabilities.
- Contributed to test and pass the 400+ testcases. achieving **EMVCo 3-D Secure (3DS) certification** for card-based transactions and supported SDK integration into **IRCTC's RailOne application**.

BHIM UPI Plugin SDK :

- Built the **foundational architecture** of the BHIM UPI Plugin SDK and delivered the complete **Widget-based and QR-based payment journey** end-to-end.
- Developed **Profile flow and widget-exposed payment flows**, integrating **NPCI's Common Library (CL) APIs** for native UPI functionality.
- Enable the merchants to integrate the sdk which enables **Native in-app UPI experience, frictionless payment, faster checkout via intent + QR** flexibility and take a role to integrate the SDK for **IOCL** (Indian Oil) and **MMRD** application.

NetBanking 2.0 SDK and Admin Portal :

- Build the **NetBanking 2.0 SDK**, enabling bank apps like **IDFC and HDFC** bank to invoke seamless, embedded transactions via SDK integration.
- Developed dual **intent-based and QR-based** payment flows with **OTP authentication** and live bank-side payment callbacks.
- Designed and developed **NetBanking 2.0 Admin portal** with **Java 17 / Spring Boot microservices and REST APIs**, along with **Spring Batch jobs, schedulers, and Kafka-based event-driven pipelines** to handle high-volume transaction ingestion and processing across a distributed system.
- Own the Ledger System module, **ledger postings, reconciliation logic, and data formatting for IBMB** forming the financial backbone of the platform.
- Optimized Oracle database performance by tuning **SQL queries, indexes, execution plans, and stored procedures**, reducing critical query execution times from minutes to milliseconds.
- Built a **Kafka-driven** worker engine using **Apache POI** to convert Excel files with **1M+ (10 lakh)** rows into a bank-specified (IBMB) format, enabling large-scale, automated file transformation for bank-side integration.

Software Developer Associate (Remote) — Mantis Technologies Private Limited *Aug 2023 – Jul 2024*

Tech Stack: React, Node.js, PostgreSQL

- Contributed to a React-based UI migration for the **Kerala RTC online ticket booking platform**, improving performance and user experience.
- Participated in PostgreSQL database design and **planning for bus stand and service managemen**t, and built mock REST APIs for early-stage integration testing.
- Took part in integrating the **BillDesk payment gateway** into the checkout and payment flow.

TECHNICAL SKILLS

Mobile: Kotlin, Java, Android SDK, Jetpack Compose, MVVM, ProGuard,Koin, Retrofit, Intent,Push Notification,Firebase,work manager
Payments & Security: UPI (NPCI Common Library), EMVCo 3DS, Tokenization, SSL Pinning, Root / Tamper Detection
Backend & Data: Spring Boot, Spring Batch, Microservices, REST APIs, Apache Kafka, Oracle (Performance Tuning), PostgreSQL, MongoDB, SQL, Node.js, Express
Frontend & Languages: React.js, Vue.js, TypeScript, JavaScript, Python
AI & Tools: LangChain, Deep agent, RAG, MCP server, Git, Docker, Linux

PROJECTS

Hostel Finder Platform — Pocket PG | Find My Hostel (findmyhostel.in)

- Built a scalable full-stack hostel booking system using Next.js, Node.js, and MongoDB with high-performance REST APIs for real-time booking.
- Implemented **AI-driven sentiment analysis** to detect fake or misleading reviews, improving recommendation accuracy.
- Designed optimized database schemas; contributed UI enhancements, map integration, and data modelling.

Blue - Deep Agent Platform | Python, LangChain, LangGraph, Streamlit

- Built a multi-agent orchestration platform where a central agent dynamically **spawns and coordinates sub-agents** for complex, multi-step tasks.

No-Sudo Software Installer (CLI-Based)

- Built a secure CLI tool with backend-driven command execution and role-based access control, enabling **developers to install whitelisted software without sudo access.**

EDUCATION

Christ College of Engineering, Irinjalakuda, Kerala (APJ Abdul Kalam Technological University) *2020 – 2024*

B. Tech in Computer Science & Engineering — CGPA: 9.45/10